

System Design Document
Airport Baggage Handling Data System

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1 Introduction

This document presents the design of a data-driven system for analyzing an airport baggage handling dataset. The dataset consists of sensor-based event logs capturing baggage movement across different processes such as check-in, screening, sorting, and transport.

The goal of the system is to provide visibility into operational performance, detect inefficiencies, and support decision-making through data analysis.

2 Data Sources

The system is based on the following sources:

- Dataset Description Document (system context, variables, and meaning) [1]
- Baggage Handling Dataset (CSV file with event logs and sensor data)[2]

The dataset contains time-series records with operational and environmental variables such as:

- Bag ID
- Timestamp
- Process (check-in, screening, sorting, transport)
- Sensor readings (temperature, vibration, speed)
- RFID Reading
- Result (success/failure)
- Priority flag
- Screen Delay time
- Vehicle Delay time

3 System Overview

The system transforms raw sensor data into insights through the following pipeline:

1. Data ingestion from CSV
2. Data cleaning and preprocessing
3. Storage in structured format
4. Analytical processing
5. Visualization and reporting

4 Entity-Relationship Diagram

Figure 1 outlines the database structure for a tracking system, likely for luggage or cargo in a transit setting.

The diagram illustrates how four main entities interact:

Bag: Stores information about individual pieces of luggage, including a unique ID, flight information (ID, terminal, zone), and whether it has priority status (1,0).

Event: Tracks specific occurrences related to a bag, sensor reading (temperature, vibration, speed), the timestamp, any delays, which process, and whether the process was a success or failure.

In summary, this schema is designed to provide a comprehensive log of a bag's journey by linking physical sensor data and status updates to specific bags and transit stages.

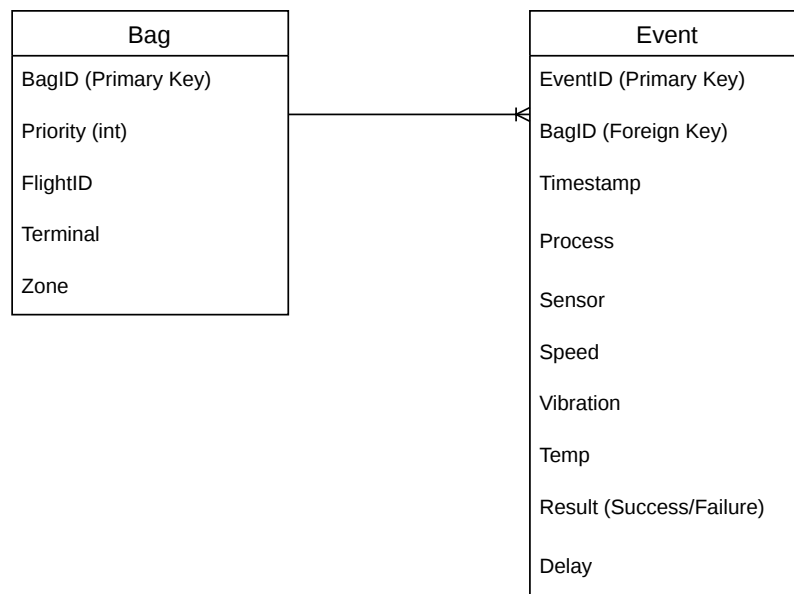


Figure 1: Entity-Relationship Diagram

5 System Architecture



Figure 2: System Architecture

6 Workflows

6.1 Data Pipeline Workflow

1. Load the CSV dataset
2. Clean missing or inconsistent values
3. Transform timestamps and categorical variables
4. Store structured data
5. Compute KPIs (delay, throughput, failure rate)

6.2 Baggage Handling System Flow

[1]



Figure 3: Baggage Handling System Workflow

7 Analytics and KPIs

The system computes the following metrics:

- Hourly Throughput (peak hours)
- Bottleneck identification (process that has the highest delay, jams, manual intervention)
- Average delay per stage
- Sensor anomaly detection
- Failure rate per stage
- Priority handling performance

8 Design Considerations

8.1 Scalability

The system should support large datasets and potential real-time extensions.

8.2 Reliability

Sensor failures and missing data must be handled accordingly.

8.3 Data Quality

Data validation and cleaning are essential due to noisy sensor inputs.

8.4 Performance

Efficient querying is required for timely analysis.

9 The Dashboard

Figure 4 illustrates a sample dashboard design for the Airport Baggage Control Center which monitors the progression of luggage through check-in, screening, sorting, and transport. The system calculates hourly throughput to identify peak hours and tracks the overall success and failure rates on every process. It identifies operational bottlenecks and jams specifically within terminals T1, T2, and T3. Priority handling performance is also measured to ensure high priority bags are processed effectively.

This comprehensive monitoring allows the center to maintain efficient baggage flow for both regular and priority bags and will support efficient and effective decision making.

10 Conclusion

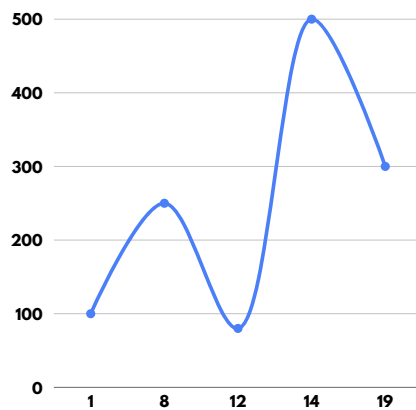
This document presents a system design for analyzing baggage handling operations using sensor data. By structuring the dataset into a clear data model and implementing analytical workflows with easy to understand dashboard and reporting, the system enables identification of inefficiencies and supports operational improvements.

Airport Baggage Control Center

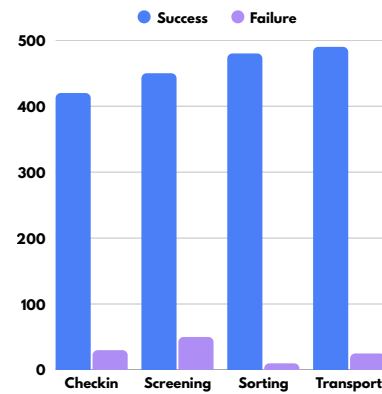
T1, T2, T3



Hourly Throughput (bags processed per hour)



Success and Failure Rate



Bottleneck identification (terminal, jams, etc)



Priority Handling Performance

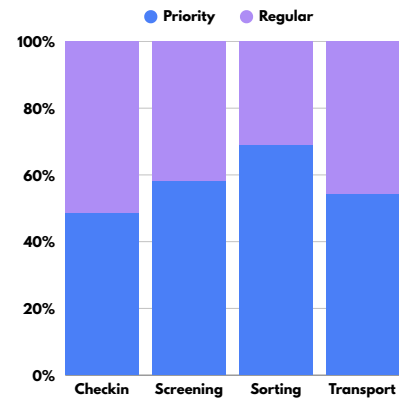


Figure 4: Dashboard Design

References

- [1] *Airport Data Insights – Dataset Description Document*. data.docx
- [2] *Baggage Handling Dataset (CSV)*.dataset.csv